Introducing while Loops

Repeat until a condition changes: counters, input checks and sentinels

Anatomy of a while loop

```
count = 1                # 1. initialise
while count <= 3:        # 2. test before every pass
    print(count)         # 3. body
    count += 1           # 4. update, or it never ends
```

OUTPUT

```
1
2
3
```

for or while?

Prefer `for`	Prefer `while`
a collection to walk through	unknown number of passes
a known count: `range(n)`	waiting for input or an event
simple, safe termination	retries, polling, game loops

Counting down

```
n = 3
while n > 0:
    print(n, end=" ")
    n -= 1
print("Liftoff!")
```

OUTPUT

```
3 2 1 Liftoff!
```

Loop until a goal is met

How many years until 1000 grows to 2000 at 7% a year?

```
balance, years = 1000, 0
while balance < 2000:
    balance *= 1.07
    years += 1
print(years)             # → 11
```

Input validation

```
age = input("Age: ")
while not age.isdigit():
    age = input("Digits only. Age: ")
print("Age is", int(age))
```

OUTPUT

```
Age is 25
```

Sentinel loop

Keep reading until a special value (the sentinel) appears.

```
total = 0
n = int(input("Number (0 stops): "))
while n != 0:
    total += n
    n = int(input("Number (0 stops): "))
print("Total:", total)
```

OUTPUT

```
Total: 15
```

while True + break

The standard pattern for menus, retries and servers.

```
while True:
    cmd = input("> ")
    if cmd == "quit":
        break
    print("You typed", cmd)
```

OUTPUT

```
You typed hi
```

Compound conditions

```
tries, ok = 0, False
while tries < 3 and not ok:
    tries += 1
    ok = tries == 2      # pretend try 2 works
print(tries, ok)
```

OUTPUT

```
2 True
```

Why does it never stop?

Cause	Fix
forgot `n -= 1` or `i += 1`	update inside the body
update moves the wrong way	down with `-=`, up with `+=`
`while x != 10` stepping by 3	use `<` or `<=`
floats: `while x != 1.0`	use `<` or `math.isclose`

REMEMBER

- Every `while` needs three things: a starting value, a test, and an update that moves toward ending.
- The test runs **before** each pass. If it is false at the start, the body never runs.
- `while True:` plus `break` is the standard pattern for menus, retries and servers.
- Prefer `<` / `<=` over `!=` when stepping, so you cannot jump past the target.
- Stuck in a loop? Press Ctrl+C to raise `KeyboardInterrupt`.
- If you can write it as `for ... in ...`, do: it cannot run forever by accident.